

Laboratorio GameShell navigazione Linux

Corso: CyberSecurity

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Esercizio: GameShell comandi di navigazione

Traccia: installare e avviare GameShell in Kali Linux, quindi completare le prime missioni usando i comandi fondamentali per muoversi tra cartelle e verificare la posizione nel file system.

Per prima cosa ho installato i pacchetti richiesti, tra cui gettext, man-db, procps, psmisc, nano, tree, bsdmainutils, x11-apps e wget. Il terminale ha completato l'installazione delle dipendenze senza errori.

```
(kali㉿kali)-[~]
└─$ sudo apt install gettext man-db procs psmisc nano tree bsdmainutils x11-apps wget
man-db is already the newest version (2.13.1-1).
procs is already the newest version (2:4.0.6-3).
psmisc is already the newest version (23.7-2).
psmisc set to manually installed.
nano is already the newest version (9.2-1).
tree is already the newest version (2.3.2-1).
tree set to manually installed.
x11-apps is already the newest version (7.7+11+b1).
x11-apps set to manually installed.
wget is already the newest version (1.25.0-3).
Installing:
  bsdmainutils  gettext

Installing dependencies:
  ncal

Suggested packages:
  calendar mailutils  gettext-doc  libgettextpo-dev
  vacation autpoint  libasprintf-dev  gnulib-l10n

Summary:
  Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 5
  Download size: 2,684 kB
  Space needed: 21.0 MB / 58.4 GB available

Continue? [Y/n] y
Get:1 http://kali.download/kali kali-rolling/main amd64 ncal amd64 12.1.8 [19.7 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 bsdmainutils all 12.1.8 [5,952 B]
Get:3 http://kali.download/kali kali-rolling/main amd64 gettext amd64 1.0-3 [2,658 kB]
Fetched 2,684 kB in 16s (172 kB/s)
Selecting previously unselected package ncal.
(Reading database... 440802 files and directories currently installed.)
Preparing to unpack ../archives/ncal_12.1.8_amd64.deb...
Unpacking ncal (12.1.8)...
Selecting previously unselected package bsdmainutils.
Preparing to unpack ../bsdmainutils_12.1.8_all.deb...
Unpacking bsdmainutils (12.1.8)...
Selecting previously unselected package gettext.
Preparing to unpack ../gettext_1.0-3_amd64.deb...
Unpacking gettext (1.0-3)...
Setting up gettext (1.0-3)...
Setting up ncal (12.1.8)...
Setting up bsdmainutils (12.1.8)...
Processing triggers for libc-bin (2.43-4)...
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2026.3.2)...
```

[illegible]

kali@kali: ~

Session Actions Edit View Help

GAMES

Press Enter to continue.

La schermata di benvenuto spiega che GameShell insegna le operazioni di base della shell: creare directory, visualizzare e modificare file, cercare contenuti e avviare programmi. Nel gioco gli oggetti e i luoghi rappresentano file e cartelle.

```
kali@kali: ~
Session Actions Edit View Help

Welcome to GameShell!

Using and administering a computer running a Unix-like operating system (e.g., Linux, BSD or macOS) requires using a command line interface called "shell". In this game you will learn how to use the shell by journeying through various missions that will teach you to perform standard tasks such as:
- creating directories,
- creating, viewing and editing files,
- searching for files satisfying certain criteria,
- running and interrupting programs,
- etc.

During your adventure you will need to visit locations, interact with people, and find various objects. However, you have to keep in mind that these are standard directories, programs, and files (often with random contents). So, for example:
- "build a hut" means creating a directory called "Hut", and
- "put a coin in your chest" means moving a file called "coin" to a directory called "Chest".

Press Enter to continue.

KALI
```

La prima missione richiedeva di raggiungere la cima della torre principale del castello. Per orientarmi sono stati utili `cd` per spostarmi, `pwd` per vedere il percorso corrente, `ls` per elencare le posizioni disponibili e `gsh check` per validare il risultato.

```

Mission goal
Go to the top of the main tower of the castle.

Useful commands

cd LOCATION
Move to the given location.
Remark: 'cd' is an abbreviation for "change directory".

pwd
Show the path to your current location.
Remark: 'pwd' is an abbreviation for "print working directory".

ls
Show a list of locations that are currently accessible.
Remark: 'ls' is an abbreviation of "list".

gsh check
Check if the mission objective has been achieved.

gsh reset
Restart the mission from the beginning.

Remarks

UPPERCASE words appearing in commands are meta-variables: you need to replace them by appropriate (string) values.

Most filesystems treat uppercase and lowercase characters differently. Make sure you use the correct path.

~
~
~
```

Ho iniziato dalla directory /home/kali/gameshell/World. Con ls ho visualizzato le aree disponibili e sono entrata in Castle. Da qui ho seguito il percorso Main_tower, First_floor, Second_floor e Top_of_the_tower, verificando ogni volta le cartelle disponibili con ls.

```
[use 'gsh help' to get a list of available commands]
[mission 1] $ 
::1          ff02::1          ff02::2          ip6-allnodes  ip6-
[mission 1] $ pwd
/home/kali/gameshell/World

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
Castle Forest Garden Mountain Stall

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Castle

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
Cellar Great_hall Main_building Main_tower Observatory

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Main_tower

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
First_floor

[use 'gsh help' to get a list of available commands]
[mission 1] $ 
::1          ff02::1          ff02::2          ip6-allnodes  ip6-
[mission 1] $ cd First_floor

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
Second_floor

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Second_floor

[use 'gsh help' to get a list of available commands]
[mission 1] $ ls
Top_of_the_tower

[use 'gsh help' to get a list of available commands]
[mission 1] $ cd Top_of_the_tower

[use 'gsh help' to get a list of available commands]
[mission 1] $ gsh check

Congratulations, mission 1 has been successfully completed!

[ progress was saved in /home/kali/gameshell-save.sh ]

|                                     |
--+-----+-----+-----+-----+
| Use the command                    |
| $ gsh help                        |
| to get the list of "gsh" commands. |
--+-----+-----+-----+
|                                     |

[mission 2] $ █
```

Il comando gsh check ha confermato il completamento della prima missione. Il progresso è stato salvato nel file /home/kali/gameshell-save.sh, così da poter riprendere l'esercizio in seguito.

La seconda missione aveva come obiettivo la cantina del castello e richiedeva di comprendere la differenza tra `cd -` e `cd ...`. Il comando `cd -` torna alla posizione precedente, mentre `cd ..` sale di un livello nella gerarchia delle directory.

```
[mission 2] $ gsh goal

{8\/_\}
\_/_\}

Mission goal
=====

Go the castle's cellar.

Secondary objective
=====

Understand the difference between ``cd -`` and ``cd ..``.

Useful commands
=====

cd -
  Jump back to the location you were in prior to your last move.

cd ..
  Move to the parent directory (one step back along the path to your current location).

pwd
  See the path to your current location.

{8\/_\}
\_/_\}

[use 'gsh help' to get a list of available commands]
[mission 2] $
```

Dalla cima della torre ho usato pwd per controllare il percorso, poi cd - per rientrare nella directory visitata prima. Ho quindi usato un percorso relativo con più livelli, ../../../../Cellar, per raggiungere la cantina.

```
[mission 2] $ pwd
/home/kali/gameshell/World/Castle/Main_tower/First_floor/Second_floor/Top_of_the_tower

[use 'gsh help' to get a list of available commands]
[mission 2] $ cd ..

[use 'gsh help' to get a list of available commands]
[mission 2] $ pwd
/home/kali/gameshell/World/Castle/Main_tower/First_floor/Second_floor

[use 'gsh help' to get a list of available commands]
[mission 2] $ cd ../../../../Cellar

[use 'gsh help' to get a list of available commands]
[mission 2] $ gsh check

Congratulations, mission 2 has been successfully completed!

[ progress was saved in /home/kali/gameshell-save.sh ]

+-----+
| Use the command |
|   $ gsh help   |
| to get the list of "gsh" commands. |
+-----+

Visual Studio
[mission 3] $ █
```

Anche in questo caso gsh check ha restituito il messaggio di completamento e GameShell ha salvato il progresso della seconda missione.

La terza missione richiedeva di tornare alla posizione iniziale e poi di arrivare nella Throne_room con soli due comandi. Era quindi necessario usare cd senza argomenti per tornare alla posizione iniziale e un percorso composto da più directory per spostarsi direttamente alla stanza del trono.

```
[mission 3] $ gsh goal

Mission goal
Go back to the starting location and then go to the throne room using only two commands.

Remark
You may experiment with as many commands as you want, but
to validate the mission the following conditions need to be met:
- the second to last command takes you to the starting point,
- the last command takes you directly to the throne room.

Useful commands
cd
Move back to GameShell's starting location.
cd LOCATION1/LOCATION2/LOCATION3
Make several moves in one command.

Remark
UPPERCASE words appearing in commands are meta-variables: you need to replace them by appropriate (string) values.

[use 'gsh help' to get a list of available commands]
[mission 3] $
```

Durante il primo controllo la missione non è stata convalidata: GameShell ha indicato che il primo comando non rispettava la condizione richiesta e mi ha riportato nella Cellar. Ho usato pwd per verificare la nuova posizione prima di correggere il percorso.

```
[mission 3] $ cd
[use 'gsh help' to get a list of available commands]
[mission 3] $ pwd
/home/kali/gameshell/World

[use 'gsh help' to get a list of available commands]
[mission 3] $ ls
Castle Forest Garden Mountain Stall

[use 'gsh help' to get a list of available commands]
[mission 3] $ ls Castle
Cellar Great_hall Main_building Main_tower Observatory

[use 'gsh help' to get a list of available commands]
[mission 3] $ ls Castle/Main_building
Library Throne_room

[use 'gsh help' to get a list of available commands]
[mission 3] $ cd Castle/Main_building/Throne_room

[use 'gsh help' to get a list of available commands]
[mission 3] $ gsh check
The previous to last command must take you to the starting point.

Sorry, mission 3 hasn't been completed.

[ progress was saved in /home/kali/gameshell-save.sh ]
You have been teleported to .../World/Castle/Cellar

+-----+
| Use the command |
| $ gsh help      |
| to get the list of "gsh" commands. |
+-----+

[mission 3] $ pwd
/home/kali/gameshell/World/Castle/Cellar

[use 'gsh help' to get a list of available commands]
[mission 3] $ cd

[use 'gsh help' to get a list of available commands]
[mission 3] $ cd Castle/Main_building/Throne_room

[use 'gsh help' to get a list of available commands]
[mission 3] $ gsh check

Congratulations, mission 3 has been successfully completed!

Well done!!!
From now on, the current location will be shown just before the command prompt.

[ progress was saved in /home/kali/gameshell-save.sh ]

+-----+
| Use the command |
| $ gsh help      |
| to get the list of "gsh" commands. |
+-----+
```

Dalla Cellar ho raggiunto la Throne_room con `cd ../Main_building/Throne_room`. Il messaggio `Congratulations mission 3 has been successfully completed` conferma la riuscita della missione e l'apprendimento dei percorsi relativi.

In conclusione, il laboratorio mi ha permesso di esercitarmi concretamente con `ls`, `pwd`, `cd`, `cd ..`, `cd -` e con i percorsi relativi. GameShell rende visibile la struttura delle directory e aiuta a comprendere come ci si muove nel file system Linux usando il terminale.